Sean Canterbury

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Backend Software Engineer | Java / Spring Boot

Backend software engineer building and maintaining **enterprise-scale inventory and logistics systems** in a distributed microservices environment. Strong focus on **test-driven development, CI/CD automation, and production reliability.** Experienced in supporting mission-critical services used in thousands of retail locations.

Technical Skills

Languages: Java, TypeScript, SQL, Python
Backend: Spring Boot, Kafka, REST APIs, Microservices
Databases: PostgreSQL, H2
Frontend: React
Testing: JUnit, Jest, Mockito, Test-Driven Development, Testkube, Automated API workflow testing, Cypress
DevOps / Tooling: Git, CI/CD Pipelines, Snyk, Linux
Practices: Agile/Scrum, Distributed Systems, Production Support, System Design

Experience

Associate Software Engineer - Lowe's Home Improvement - June 2025 - Present
Backend Engineer on the Store Inventory Management System, supporting shelf replenishment, packout, and freight workflows across large-scale retail operations.

- Led integration of **automated API and regression testing** into CI/CD pipelines using Testkube, enabling tests to run automatically on release deployments.
- Designed, enhanced, and maintained **Spring Boot Microservices** powering store inventory, replenishment, and freight workflows
- Integrated **Kafka event producers/consumers** to enable asynchronous inventory updates
- Increased **unit test coverage by 10-15%** across core services, improving deployment confidence and reducing production defects
- Developed and validated REST APIs handling **product delivery workflows and inventory state transitions**
- Provided routine **production support and on-call assistance**, diagnosing and resolving issues impacting live store operations
- Collaborated cross-functionally with frontend teams to validate backend changes, stage releases, and support feature demos to business stakeholders.
- Contributed to security and reliability initiatives, including **Snyk vulnerability remediation** and automated deployment workflows.

Tech: Java, Spring Boot, Kafka, PostgreSQL, CI/CD, Automated Testing, REST APIs, Testkube

Associate Software Engineer - Launchpad Program (Internal) Lowe's Home Improvement - March 2025 - June 2025
Completed an intensive internal engineering program emphasizing test-driven development and full-stack application design.

- Built a full-stack **order scheduling web application** using **Spring Boot, React, and SQL**
- Delivered a complete MVP within a **2-week capstone sprint,** meeting all functional and quality requirements
- Applied **strict TDD practices,** achieving **100% unit test coverage** across both backend and frontend services.
- Designed features supporting personal and public projects, order placement, delivery scheduling, item lifecycle management, and cost calculations.
- Implemented role-aware behavior for delivered vs pending inventory items and project order state transitions

Tech: Java, Spring Boot, React, TypeScript, SQL, H2 Database, JUnit, Jest, Cypress

Software Quality Engineer Intern - Gilero - September 2024 - March 2025

- Performed software verification and validation for medical device systems in compliance with **IEC 62304** and **ISO 14971**
- Conducted **risk analysis**, integration testing, and unit testing across multiple software tools.
- Led execution of **CAPA procedures** to address compliance-critical issues in regulated environments.

Education

University of North Carolina at Charlotte, Charlotte, NC | August 2020 - May 2024
Bachelor of Science in Computer Science | Concentration: Software, Systems, and Networking | Minor: Mathematics GPA: 3.5
Honors: Chancellor's List: Spring 2024, Fall 2023 | Dean's List: Spring 2023, Fall 2022, Fall 2020